

Problem Description

The Monk recently learnt about priority queues and requested his teacher for a fun challenge to solve. As a result, his teacher devised a simple task. He now possesses A , which is an integer array. He wishes to discover the product of the greatest, second largest, and third largest integers in the range $[1, i]$ for each index i .

Note: Two numbers can be the same value-wise but they should be distinct index-wise.

Constraints:

$$1 \leq N \leq 100000$$

$$0 \leq A[i] \leq 1000000$$

Input:

The first line contains an integer N , denoting the number of elements in the array A .

The next line contains N space separated integers, each denoting the i th integer of the array A .

Output:

Print the answer for each index in each line. If there is no second largest or third largest number in the array A upto that index, then print "-1", without the quotes.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

5
1 2 3 4 5

EXPECTED OUTPUT

-1
-1
6
24
60

Test Case 2

INPUT (STDIN)

8
3 7 1 2 4 7 5 9

EXPECTED OUTPUT

-1
-1
21
42
84
196
245
441

▼ Mandatory Test Cases

Test Case 1

KEYWORD

if(biggest<big)

Test Case 2

KEYWORD

if(a[i]>biggest)

▼ Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

CYCLOMATIC COMPLEXITY

10

TOKEN COUNT

300

NLOC

62

```
1 #include <stdio.h>
2
3 void findProduct(long long a[], int n) {
4     for (int i = 0; i < n; i++) {
5         if (i < 2) {
6             printf("-1\n");
7             continue;
8         }
9         long long biggest = -1, big = -1, small = -1;
10        for (int j = 0; j <= i; j++) {
11            if (a[j] > biggest) {
12                small = big;
13                big = biggest;
14                biggest = a[j];
15            } else if (a[j] > big) {
16                small = big;
17                big = a[j];
18            } else if (a[j] > small) {
19                small = a[j];
20            }
21        }
22        if (biggest < big) {
23
24        }
25        printf("%lld\n", biggest * big * small);
```

```
26     }  
27 }  
28  
29 int main() {  
30     int n;  
31     if (scanf("%d", &n) != 1) return 0;  
32     long long a[n];
```

```
33     for (int i = 0; i < n; i++) {  
34         scanf("%lld", &a[i]);  
35     }  
36     findProduct(a, n);  
37     return 0;  
38 }  
39
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

5

1 2 3 4 5

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Queue1.exe

-1

-1

6

24

60

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

8

3 7 1 2 4 7 5 9

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Queue1.exe

-1

-1

21

42

84

196

245

441

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>